Aging as Synchronization Failure: A Minimal Model Explaining the Scaling Limits of Anti-Aging Interventions

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Abstract

We present a minimal computational model of biological aging based on the dynamics of coupled oscillators. The model proposes that aging emerges primarily from progressive desynchronization among biological regulatory systems, rather than from the accumulation of component-level damage alone. A positive feedback loop—increasing variance σ raises synchronization cost, which depletes the regulatory buffer B and coupling strength K, which further increases σ—produces aging dynamics consistent with empirically observed patterns including the Gompertz mortality law. Crucially, the model generates a quantitative prediction regarding intervention efficacy: local repair of a single oscillator reduces system-wide desynchronization by approximately $\Delta\sigma/\sigma \sim (2f - f^2)/n$, where f is repair effectiveness and n is system complexity. This

scaling law implies that local anti-aging interventions, highly effective in simple systems such as mice, lose therapeutic significance in complex systems such as humans. Global synchronization interventions, by contrast, maintain constant effectiveness independent of n. The model provides a mechanistic account of the well-documented failure to translate anti-aging findings from animal models to humans.

Keywords: aging, Kuramoto oscillators, synchronization, Gompertz law, anti-aging interventions, systems biology

1 Introduction

The dominant paradigm in aging research focuses on the accumulation of molecular damage: telomere shortening, mitochondrial dysfunction, DNA error accumulation, and oxidative stress [8]. This framework has produced valuable insights but faces a persistent explanatory gap: damage accumulates approximately linearly, yet aging accelerates nonlinearly. An eighty-year-old is not simply "twice as old" as a forty-year-old in any functional sense—they occupy a qualitatively different physiological regime.

A complementary perspective focuses not on damage per se but on the loss of coordination among biological systems. With age, circadian rhythms desynchronize from metabolic rhythms; cortisol cycles lose precision; neural networks that operate coherently in youth become progressively decoupled. The variance of biological parameters—glucose levels, blood pressure, hormone concentrations—increases with age not because mean values deteriorate dramatically, but because the systems that regulate these parameters lose mutual coherence.

This observation motivates a synchronization-theoretic approach to aging. The Kuramoto model, originally developed to describe the spontaneous synchronization of coupled oscillators [1], provides a natural mathematical framework. In this framework, the biological organism is modeled as a network of oscillators—each representing a major regulatory system—that maintain coherence through coupling forces. Aging corresponds to the progressive loss of this coherence, driven by a positive feedback loop between desynchronization, energetic cost, and structural damage.

Independent empirical support for this view comes from longitudinal analysis of blood biomarkers, which reveals that the time required for organisms to return to homeostasis after perturbation increases approximately fourfold between ages 40 and 90 [2]—a signature of systems approaching a critical transition, consistent with the synchronization framework.

The present work introduces a minimal implementation of this framework, derives its key quantitative predictions, and explores their implications for the design of anti-aging interventions.

2 Model

2.1 Biological oscillators

Each organism is represented as a network of N coupled oscillators with phases $\theta_i(t)$ governed by the Kuramoto equation with noise [1, 3]:

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N}\sum_j \sin(\theta_j - \theta_i) + \eta_i(t) \tag{1}$$

where $\omega_i \sim \text{Uniform}(0.9, 1.1)$ are natural frequencies, K is coupling strength, and $\eta_i(t)$ is Gaussian noise with standard deviation σ_{noise}.

The degree of synchronization across the network is measured by the order parameter $r = |\langle e^{i\theta}\rangle|$, and desynchronization is defined as:

$$\sigma = 1 - r \in [0, 1] \tag{2}$$

where $\sigma = 0$ indicates perfect synchronization and $\sigma = 1$ indicates complete incoherence.

2.2 Damage feedback loop

The key innovation of the model is the introduction of a positive feedback loop connecting desynchronization to structural damage:

$$\text{Cost} = \frac{\sigma^{3.5}}{B} \tag{3}$$

$$\frac{dD}{dt} = \text{Cost} \tag{4}$$

$$\frac{dB}{dt} = -\alpha \cdot \text{Cost} \tag{5}$$

$$\frac{dK}{dt} = -\beta \cdot \text{Cost} \cdot (1 + \sigma) \tag{6}$$

where D is accumulated damage, B is the regulatory buffer (initial value 1.0), and α, β are degradation parameters. The nonlinear cost function ($\sigma^{3.5}$) ensures that small increases in desynchronization produce disproportionately large energetic costs at high σ, consistent with the Jensen's inequality argument for multiplicative biological systems.

Additionally, the effective noise amplitude grows exponentially with accumulated damage:

$$\eta_{\text{eff}} = \eta_0 \cdot \min\left(\exp\left(\frac{0.25 \cdot D}{D_{\text{scale}}}\right),\ 5.0\right) \tag{7}$$

This captures the empirical observation that aging systems become progressively less able to buffer perturbations, producing increasing variability in biological parameters [2].

2.3 Organism parameters

Two canonical organisms are implemented with the parameters shown in Table 1.

Table 1: Model parameters for mouse and human.

Parameter	Mouse (n=5)	Human (n=20)
Oscillators N	5	20
Coupling K_0	0.80	0.30
Noise σ_0	0.05	0.15
α	0.0015	0.0020
β	0.003	0.004

We emphasize that these mappings are illustrative rather than calibrated: the distinction between n=5 and n=20 represents a qualitative difference in regulatory complexity, not a literal count of independent biological oscillators. An operational measure of n can be defined empirically as the number of principal components explaining $\geq$90% of variance in a panel of biological markers — a quantity directly measurable in existing biobank datasets such as UK Biobank or NHANES, and one that would allow the scaling law to be tested quantitatively against real intervention data.

3 Results

3.1 Emergence of the Gompertz mortality law

The qualitative behavior of the model is robust across a wide range of parameter values (noise amplitude, coupling strength, α, β): both the emergence of Gompertz-like mortality dynamics and the $1/n$ scaling law are preserved. Furthermore, both the nonlinear cost function ($\sigma^{3.5}$) and the exponential noise feedback are necessary for reproducing Gompertz-like dynamics; removing either eliminates the observed acceleration of mortality risk and produces approximately linear damage accumulation instead.

Simulation of cohorts of 200 agents with death defined as $B < 0.02$ or $\sigma > 0.95$ produces survival curves consistent with empirically observed mortality patterns. Critically, the log-hazard $h(t)$ follows an approximately linear trend in time—the defining characteristic of the Gompertz mortality law—despite this law not being explicitly encoded in the model. This emergence occurs because the exponential noise feedback creates exponentially accelerating desynchronization near the death threshold, producing the characteristic doubling of mortality risk with age.

The median survival time differs between mouse ($t_{50} \approx 80$) and human ($t_{50} \approx 72$) in model time units. These units are dimensionless and cannot be directly compared to biological lifespan in years; the model captures the *relative* difference in aging trajectory between organisms of different complexity, not absolute longevity. The human curve shows greater late-life acceleration—consistent with the more complex coupling architecture being less recoverable once desynchronization initiates.

3.2 Complexity determines aging trajectory

Figure 1 constitutes the central empirical result of the model. It shows the desynchronization trajectory $\sigma(t)$ averaged over 15 independent simulations for networks of varying size ($n = 3, 5, 8, 12, 20, 30$). The key finding is monotonic: larger networks enter the high-

desynchronization regime earlier and more reliably. The standard deviation of σ decreases with n, indicating that complex systems, while more susceptible to desynchronization, are more predictable in their aging trajectory.

This result emerges from the interplay between coupling architecture and noise accumulation. In small networks, strong relative coupling can periodically restore coherence, producing observed returns to low σ. In large networks, the same absolute coupling represents a smaller fraction of the total synchronization demand, making recovery progressively less likely.

3.3 Scaling law for intervention efficacy

The central quantitative prediction of the model concerns the effectiveness of interventions that repair a single oscillator versus interventions that enhance global coupling. We derive this law from first principles.

Derivation

In the mean-field Kuramoto model with noise, each oscillator i has phase θ_i. Near a synchronized state, phases fluctuate around the mean-field phase Ψ with small deviations $\phi_i = \theta_i - \Psi$. Expanding the complex order parameter $Z = \frac{1}{N}\sum_j e^{i\theta_j}$ to second order:

$$r = |Z| \approx 1 - \frac{1}{2N}\sum_{j=1}^{N}\langle\phi_j^2\rangle \tag{8}$$

In the linearized dynamics, the steady-state phase variance of oscillator j is proportional to its squared noise amplitude: $\langle\phi_j^2\rangle = c \cdot \eta_j^2$, where c is a constant depending on coupling strength and frequency distribution. Therefore:

$$\sigma = 1 - r \approx \frac{c}{2N}\sum_{j=1}^{N}\eta_j^2 \tag{9}$$

Proposition. Let all oscillators have identical noise η_0, so that $\sigma_0 = \frac{c\eta_0^2}{2}$. Suppose a local intervention reduces the noise of a single oscillator k by fraction $f \in [0, 1]$, setting

$\eta_k \leftarrow (1-f)\eta_0$. Then:

$$\boxed{\frac{\Delta\sigma}{\sigma_0} = \frac{f(2-f)}{N}} \quad \text{(local)} \tag{10}$$

Proof. After the intervention:

$$\begin{aligned}\sigma_{\text{new}} &= \frac{c}{2N}\left[(N-1)\eta_0^2 + (1-f)^2\eta_0^2\right] \\ &= \frac{c\eta_0^2}{2N}\left[N - 2f + f^2\right]\end{aligned} \tag{11}$$

Therefore:

$$\Delta\sigma = \sigma_0 - \sigma_{\text{new}} = \frac{c\eta_0^2}{2N}\left[N - (N - 2f + f^2)\right] = \frac{c\eta_0^2}{2N}(2f - f^2)$$

$$\frac{\Delta\sigma}{\sigma_0} = \frac{c\eta_0^2(2f - f^2)/(2N)}{c\eta_0^2/2} = \frac{f(2-f)}{N} \qquad \square \tag{12}$$

Global intervention. If the same fractional repair f is applied to all N oscillators simultaneously:

$$\sigma_{\text{new}}^{\text{global}} = \frac{c\eta_0^2}{2}(1-f)^2 \tag{13}$$

$$\left.\frac{\Delta\sigma}{\sigma_0}\right|_{\text{global}} = 1 - (1-f)^2 = f(2-f) \tag{14}$$

This is independent of N. The ratio of local to global efficacy is therefore exactly $1/N$:

$$\frac{(\Delta\sigma/\sigma_0)_{\text{local}}}{(\Delta\sigma/\sigma_0)_{\text{global}}} = \frac{1}{N} \tag{15}$$

We note a correction to our earlier shorthand: the precise formula is $f(2-f)/N$, not f^2/N. The two expressions agree for $f \ll 1$ (small repair) and for $f = 1$ (complete repair) give $1/N$ vs $1/N$, but differ at intermediate f. Table 2 reports the exact values.

Numerical values

For $f = 0.5$ (50% noise reduction), $f(2 - f) = 0.75$:

Table 2: Exact predicted reduction $\Delta\sigma/\sigma$ for local vs. global interventions ($f = 0.5$, $f(2 - f) = 0.75$).

System (n)	Local: $\frac{f(2-f)}{n}$	Global: $f(2 - f)$
3 (simple organism)	25.0%	75.0%
5 (mouse)	15.0%	75.0%
12	6.3%	75.0%
20 (human)	3.8%	75.0%
30	2.5%	75.0%

The global intervention maintains constant efficacy regardless of system complexity. This divergence between local and global efficacy, scaling exactly as $1/N$, constitutes the primary falsifiable prediction of the model.

Robustness of the scaling

The $1/N$ ratio between local and global efficacy is exact within the mean-field approximation and does not depend on the specific values of coupling K, noise η_0, frequency distribution, or the repair fraction f. It holds as long as (i) noise contributions to the order parameter are additive, and (ii) oscillators are statistically exchangeable. The first assumption holds in the linearized regime; the second is violated in hierarchical networks (a limitation discussed in Section 4.3).

We verified numerically that the qualitative scaling is preserved when the cost exponent is varied between 2 and 5, when α and β are varied by factors of 2–5, and when n varies continuously from 3 to 30.

4 Discussion

4.1 The translational gap in anti-aging research

The failure to translate anti-aging interventions from animal models to humans is one of the most consistent and costly problems in aging research. Interventions targeting specific molecular pathways—caloric restriction mimetics, senolytics, NAD^+ precursors, mTOR inhibitors—show dramatic effects in short-lived organisms but modest or negligible effects in clinical trials [4].

The scaling law derived here provides a mechanistic explanation for this failure that requires no assumption of fundamental biological difference between species. A local intervention is inherently less effective in a more complex organism, not because the target pathway is less important, but because any single pathway represents a smaller fraction of the system's total synchronization demand. An intervention that addresses 25% of synchronization variance in a three-oscillator system (n=3, f=0.5: $f(2-f)/n = 0.75/3 = 25\%$) addresses only 3.8% in a twenty-oscillator system (n=20: $0.75/20 = 3.8\%$), at identical molecular effectiveness.

4.2 Implications for intervention design

The model suggests a reorientation of anti-aging strategy from local repair toward global synchronization. Interventions that strengthen the coupling among regulatory systems—rather than repairing any specific system—maintain efficacy independent of organism complexity. Empirically, this corresponds to interventions including regular physical exercise at consistent times, structured sleep protocols, intermittent fasting, and other zeitgeber-based approaches. These interventions are notable for producing consistent (if modest) benefits across both animal models and human clinical populations [5], consistent with complexity-independent efficacy.

This reorientation does not imply that local molecular interventions are without value.

It implies that their value is primarily diagnostic and mechanistic rather than therapeutic—they identify the components of biological synchronization networks, but the therapeutic target is the network itself.

4.3 Model limitations

The present model is deliberately minimal and makes several simplifying assumptions that warrant explicit acknowledgment.

First, all oscillators are treated as equivalent in role and coupling architecture. Biological regulatory networks have hierarchical structure—the suprachiasmatic nucleus serves as a master oscillator—and this hierarchy is not captured in the current implementation.

Second, the mapping between model parameters and biological quantities is illustrative rather than calibrated.

Third, the model does not account for developmental processes, genetic variation, or environmental heterogeneity that substantially influence aging in both animal models and human populations.

These limitations suggest directions for model extension rather than fundamental challenges to the core framework. The model should be interpreted as a minimal (toy) model capturing qualitative system-level dynamics, rather than a quantitatively calibrated physiological simulator.

4.4 Testable predictions

The model generates three predictions testable with existing methods:

Prediction 1 (primary): The effectiveness of single-pathway interventions in reducing biological age markers scales approximately as $1/n$, where n is a measure of regulatory network complexity.

Prediction 2: The cross-correlational structure of biological parameters—the degree to which circadian, metabolic, immune, and hormonal rhythms are mutually correlated—should

be a better predictor of functional age than the mean values of any single parameter.

Prediction 3: Zeitgeber-based interventions should show greater relative benefit in older organisms with high existing desynchronization than in younger organisms with intact synchronization, because the marginal value of coupling restoration is highest near the desynchronization threshold.

5 Conclusion

We have presented a minimal model in which aging emerges as a positive feedback loop of desynchronization in coupled biological oscillators. The model reproduces the Gompertz mortality law without encoding it directly, generates differential aging trajectories as a function of system complexity, and produces the quantitative scaling law $\Delta\sigma/\sigma = (2f - f^2)/n$ for local repair interventions.

The primary implication is practical: the consistent failure to translate anti-aging interventions from simple organisms to humans is not a biological mystery requiring species-specific explanation—it is a mathematical consequence of the scaling properties of synchronization dynamics. Interventions targeting specific components of a synchronization network necessarily lose efficacy as network complexity grows. Interventions targeting the network coupling itself do not.

Aging, in this framework, is not primarily a story of damage accumulation. It is a story of a network gradually losing its ability to coordinate. The most direct therapeutic target is the coordination itself.

This suggests that aging interventions should be evaluated not only by their local biochemical effects, but by their impact on system-wide synchronization—a criterion largely absent from current clinical trial design.

Code availability

Simulation code will be made available upon publication. Contact: `piotrczary@gmail.com`

Competing interests

The author declares no competing interests.

A Scaling of Interventions in the Kuramoto Framework

A.1 Variance approximation of desynchronization

Consider a system of n coupled oscillators with phases θ_i. The Kuramoto order parameter is:

$$r = \left| \frac{1}{n} \sum_{i=1}^{n} e^{i\theta_i} \right| \tag{16}$$

We define macroscopic desynchronization as $\sigma = 1 - r$. For a synchronized state with small phase dispersion, Taylor expansion yields $e^{i\theta_i} \approx 1 + i\theta_i - \theta_i^2/2$. Assuming the mean phase is centered ($\langle \theta_i \rangle = 0$), the real part of the order parameter is approximately:

$$r \approx 1 - \frac{1}{2} \operatorname{Var}(\theta) \tag{17}$$

Thus, system desynchronization scales directly with phase variance: $\sigma \approx \frac{1}{2} \operatorname{Var}(\theta)$.

A.2 Noise-driven variance decomposition

Assuming each oscillator is perturbed by independent noise ξ_i with amplitude (standard deviation) η_i, the steady-state phase variance scales as $\langle \phi_i^2 \rangle = c\, \eta_i^2$, where c depends on coupling

strength and frequency distribution. The total system desynchronization is therefore:

$$\sigma = \frac{c}{2n} \sum_{i=1}^{n} \eta_i^2 \tag{18}$$

For uniform baseline noise $\eta_i = \eta_0$: $\sigma_0 = \frac{c\,\eta_0^2}{2}$.

A.3 Local vs. global interventions

Let an intervention reduce the noise amplitude of target oscillator k by fraction $f \in (0, 1]$, so $\eta_k \to (1 - f)\eta_k$.

The change in variance contribution of oscillator k is:

$$\Delta(\eta_k^2) = \eta_0^2 - (1 - f)^2 \eta_0^2 = \eta_0^2 (2f - f^2) \tag{19}$$

For a **local intervention** affecting only one oscillator out of n:

$$\frac{\Delta\sigma}{\sigma_0} = \frac{c\,\eta_0^2 (2f - f^2)/(2n)}{c\,\eta_0^2/2} = \frac{2f - f^2}{n} \tag{20}$$

For a **global intervention** where all n oscillators receive the same fractional improvement f:

$$\frac{\Delta\sigma}{\sigma_0} = 1 - (1 - f)^2 = 2f - f^2 \tag{21}$$

This is independent of n.

A.4 Conclusion

The ratio of local to global efficacy is exactly $1/n$, independent of f, η_0, coupling strength, or frequency distribution:

$$\frac{(\Delta\sigma/\sigma_0)_{\text{local}}}{(\Delta\sigma/\sigma_0)_{\text{global}}} = \frac{1}{n} \tag{22}$$

For a complete local repair ($f = 1$): $(2f - f^2)|_{f=1} = 1$, so local efficacy reduces precisely

to $1/n$. Even a hypothetically perfect intervention eliminating all noise in a single oscillator is bounded by network complexity. This structural property holds for any $n \geq 1$ and any $f \in (0, 1]$.

A.5 Regime of validity

The derivation above relies on two approximations that warrant explicit acknowledgment.

First, the Taylor expansion $r \approx 1 - \frac{1}{2}\mathrm{Var}(\theta)$ holds only for *small phase dispersion*. In the main simulations, σ reaches values approaching 0.95 — near-complete desynchronization — where this linearization is quantitatively inaccurate. The analytical derivation therefore rigorously establishes the $1/n$ scaling in the weakly-synchronized regime. Numerical simulations confirm that the scaling generalizes beyond this regime (Section 3.3), but we cannot claim analytical proof in the high-desynchronization limit.

Second, the variance decomposition treats oscillator phases as independent. However, the Kuramoto coupling term explicitly introduces phase correlations. The derivation ignores coupling-induced covariances, which is justified in the limit $K \to 0$ or when phase fluctuations are small relative to 2π.

In summary: the $1/n$ scaling law is proven exactly within the weak-coupling, small-dispersion approximation. Numerical simulations confirm that the qualitative result persists across the full dynamical range of the model, including the high-σ regime where system failure occurs.

B Robustness of Gompertz Dynamics to Cost Function Form

B.1 Objective

The baseline model employs a cost function $C(\sigma) \propto \sigma^x$ with $x = 3.5$. This appendix demonstrates that Gompertz-like mortality dynamics emerge for any $x > 1$, depending only on two structural properties: (i) convexity of the cost function, and (ii) positive feedback between accumulated damage and noise amplitude.

B.2 Generalized feedback structure

Consider the system:

$$\frac{dD}{dt} = \frac{\sigma^x}{B} \tag{23}$$

$$\frac{dB}{dt} = -\alpha \frac{\sigma^x}{B} \tag{24}$$

$$\frac{dK}{dt} = -\beta \frac{\sigma^x}{B}(1+\sigma) \tag{25}$$

with exponential noise scaling $\eta_{\text{eff}} = \eta_0 \cdot \exp(\gamma D)$, $\gamma > 0$. The system contains a closed positive feedback loop:

$$\sigma \uparrow \Rightarrow \; C \uparrow \Rightarrow \; D \uparrow \Rightarrow \; \eta \uparrow \Rightarrow \; \sigma \uparrow \tag{26}$$

B.3 Late-stage approximation and finite-time singularity

Near the end of life, B is small and noise dominates coupling. We approximate $dD/dt \sim \sigma^x$. This approximation assumes σ is dominated by noise rather than coupling ($K \approx 0$ regime), which is justified late in the simulation when K has been depleted by accumulated damage.

Due to exponential noise amplification, $\sigma(t) \sim \exp(\gamma D(t))$, giving:

$$\frac{dD}{dt} \sim e^{x\gamma D} \tag{27}$$

Separating variables and integrating:

$$D(t) \sim -\frac{1}{x\gamma} \log(t_c - t) \tag{28}$$

for some critical time t_c, showing $D \to \infty$ as $t \to t_c$ — a finite-time singularity characteristic of systems undergoing catastrophic failure.

B.4 Implication for mortality hazard

Since $\sigma(t) \sim \exp(\gamma D(t))$ accelerates toward the failure threshold, the instantaneous hazard in the pre-critical regime satisfies:

$$h(t) \propto \exp(\lambda t), \quad \lambda > 0 \tag{29}$$

This is the Gompertz law: $\log h(t)$ grows linearly in time. The rate λ depends on x and the degradation parameters, but the *qualitative structure* is preserved for all $x > 1$.

B.5 Independence from exponent x

The exponent x influences the rate of acceleration but not the form of the dynamics. For any $x > 1$: the cost is convex, the feedback loop is active, damage accelerates super-exponentially, and hazard grows approximately exponentially. Gompertz-like dynamics therefore emerge from the architecture of the model, not from parameter tuning.

Key result: Gompertz-like dynamics emerge for all $x > 1$.

B.6 Numerical verification

We simulated the model with cost exponents $x \in \{2.0, 2.5, 3.0, 3.5, 4.0, 5.0\}$. In all cases, the log-hazard curve shows approximately linear growth in the late-life regime, confirming that Gompertz emergence is a structural property of the desynchronization feedback. The exponent $x = 3.5$ was selected because it produces a biologically plausible lifespan distribution with a clear plateau before rapid late-life decline; exponents in the range $[2.5, 4.5]$ produce qualitatively identical results.

Conclusion: Aging acceleration in this model emerges from system architecture — convex cost and feedback-amplified noise — not from the choice of x.

C Scaling Law Under Non-Mean-Field Topologies

C.1 Motivation

The scaling law $(\Delta\sigma/\sigma)_{\text{local}}/(\Delta\sigma/\sigma)_{\text{global}} = 1/n$ was derived under the mean-field assumption: all oscillators are coupled equally to all others. Real biological regulatory networks are not fully connected — they have hierarchical structure, hub nodes (e.g., the suprachiasmatic nucleus as master clock), and modular organization. This appendix examines how topology affects the $1/n$ scaling.

C.2 Three topological regimes

We consider three canonical topologies:

(i) Mean-field (all-to-all): Every oscillator couples to every other with equal weight K/n. This is the baseline case proven in Appendix A.

(ii) Hub-and-spoke: One master oscillator (hub) couples to all others; peripheral oscillators do not couple to each other. This approximates the biological circadian hierarchy, where the suprachiasmatic nucleus entrains peripheral clocks.

(iii) Modular: Oscillators are partitioned into m modules of size n/m, with strong within-module coupling and weak between-module coupling. This approximates the organization of major physiological systems (cardiovascular, metabolic, immune, etc.) that couple loosely to each other.

C.3 Effect on local intervention efficacy

The key question is: does a local repair of one oscillator produce a $1/n$ effect, or does topology create amplification or suppression?

Hub-and-spoke: Repairing the hub oscillator affects all n spokes through the coupling term — the effect is $O(1)$, not $O(1/n)$. Repairing a single spoke affects only that spoke's contribution to the order parameter, giving $O(1/n)$ as in the mean-field case. This creates a *heterogeneous* intervention landscape: hub repair scales as a global intervention, spoke repair scales locally.

Modular topology: Repairing one oscillator within a module of size n/m produces a local effect of order m/n — larger than the mean-field $1/n$ by a factor of m. However, this improvement is *confined to the module*: since between-module coupling is weak, the repaired module does not pull the rest of the system toward synchrony. The global desynchronization σ is dominated by between-module variance, which the local repair does not address.

C.4 General bound

Across all topologies, the following bound holds:

$$\frac{(\Delta\sigma/\sigma)_{\text{local}}}{(\Delta\sigma/\sigma)_{\text{global}}} \leq \frac{1}{n_{\text{eff}}} \tag{30}$$

where n_{eff} is the *effective number of independent regulatory units* — the number of oscillators that contribute independently to total system variance. For fully connected networks, $n_{\text{eff}} = n$. For modular networks, $n_{\text{eff}} \approx m$ (number of modules), since within-module oscilla-

tors are highly correlated and contribute as a single unit.

The important consequence: in modular networks, the $1/n$ scaling becomes $1/m$ — which is *more pessimistic* than mean-field for the same total n, because the effective dimensionality is lower. A human with $n = 20$ oscillators organized into $m = 6$ physiological modules would show local intervention efficacy of at most $1/6 \approx 17\%$ per module — but since each module contains multiple oscillators, repairing one oscillator within a module gives at most $1/n$ system-wide improvement.

C.5 Hub repair as global intervention

The hub-and-spoke analysis reveals a topological exception: interventions targeting high-centrality nodes (master oscillators, hub genes, master regulators) can achieve near-global efficacy from a single targeted repair. This is consistent with the empirical observation that interventions targeting the circadian master clock (light therapy, chronotherapy) show broader physiological effects than most molecular interventions.

This does not contradict the main scaling argument. It refines it: the $1/n$ bound applies to *structurally peripheral* components. Interventions targeting *structural hubs* can exceed this bound — but the number of true hubs in any biological network is small by definition.

C.6 Conclusion

The $1/n$ scaling law is robust across topologies in the following sense: for interventions targeting non-hub, non-master components (which comprise the vast majority of molecular targets in current anti-aging research), the system-wide efficacy scales inversely with the effective number of independent regulatory units. Topology can create exceptions for high-centrality targets, but these exceptions are rare by network structure.

This analysis strengthens rather than weakens the main conclusion: the translational gap between mice and humans is not merely a consequence of total oscillator count, but of the greater modular complexity of human regulatory architecture — a complexity that

makes hub-targeting strategies potentially more valuable than pathway-targeting strategies in human aging research.

D Directional Test of Desynchronization Predictions in Population Data

D.1 Dataset

We analyzed six physiological markers — fasting glucose, total cholesterol, high-sensitivity CRP, triglycerides, and systolic and diastolic blood pressure — in NHANES 2017–2018 participants aged 20–79 (n = 2,673, after exclusion of participants with any missing marker value). Participants were divided into six decade-wide age bins (Table 3).

Table 3: Sample sizes per age bin, NHANES 2017–2018.

Age bin	n
20–29	—
30–39	—
40–49	—
50–59	—
60–69	—
70–79	—
Total	2,673

D.2 PCA and effective dimensionality

Principal component analysis on z-scored markers yields cumulative explained variance:

$$[0.300,\ 0.486,\ 0.662,\ 0.811,\ \mathbf{0.915},\ 1.000] \tag{31}$$

The number of components explaining $\geq 90\%$ of variance is $n_{\text{eff}} = 5$.

This result has an important interpretation: six metabolic markers, despite representing

nominally distinct physiological systems, share substantial covariance and are effectively spanned by 5 independent dimensions. However, this panel captures primarily metabolic regulation. A broader panel including circadian rhythm markers (melatonin, cortisol diurnal profile), immune parameters, and endocrine axes would be expected to yield higher n_{eff}, as these systems are regulated by partially independent mechanisms.

The finding that $n_{\text{eff}} = 5$ for this metabolic panel does not imply that human regulatory complexity equals that of mice — it implies that this particular marker set captures one regulatory module rather than the full hierarchy. This is consistent with the model's framework: n should be interpreted as the number of independent regulatory modules relevant to the intervention target, not total organism complexity.

D.3 Results

Within-system variability (mean standard deviation across z-scored markers) by age decade:

$$[25.6,\ 32.8,\ 37.8,\ \mathbf{48.4},\ 30.7,\ 39.6] \tag{32}$$

The general trend is increasing ($R^2 = 0.77$, $p = 0.021$; Figure **??**), consistent with model prediction of rising σ with age. The non-monotonic drop at 60–69 likely reflects survival selection: the least healthy individuals in this cohort have already died, leaving a relatively healthier survivor population.

Mean absolute cross-system correlation by age decade:

$$[0.159,\ 0.207,\ 0.157,\ 0.110,\ 0.115,\ 0.117] \tag{33}$$

A declining trend is present from the fourth decade onward ($R^2 = 0.56$, $p = 0.087$; Figure **??**), though not reaching conventional significance. The elevated value in the 30–39 bin is visible in Figure D2 and is discussed below.

D.4 Interpretation

These findings do not confirm the synchronization framework — the cross-sectional design, limited marker selection, and non-monotonic patterns prevent strong inference. They constitute a *directional test*: the model predicts that σ should increase and cross-system correlation should decrease with age. Both signals are present in the expected direction.

The $n_{\text{eff}} = 5$ result from PCA provides an empirical anchor for the model parameter. For interventions targeting this metabolic regulatory module specifically, the predicted local efficacy at $f = 0.5$ is $(2f - f^2)/n_{\text{eff}} = 0.75/5 = 15\%$ — matching the mouse parameter in the main model. This reinforces the point that what matters is not total organism complexity but the effective dimensionality of the *relevant* regulatory module.

The elevated cross-correlation in the 30–39 bin may reflect cohort composition, medication use patterns, or cross-sectional design effects. Circadian rhythm amplitude and precision are known to decline with age [6], consistent with declining cross-system correlation as a marker of regulatory decoupling.

A longitudinal dataset with repeated individual measurements (e.g., **(author?)** 2) with direct trajectory fitting would provide a substantially stronger test of the desynchronization hypothesis.

References

[1] Kuramoto, Y. (1975). Self-entrainment of a population of coupled non-linear oscillators. In *International Symposium on Mathematical Problems in Theoretical Physics*, Lecture Notes in Physics, Vol. 39, pp. 420–422. Springer, Berlin.

[2] Pyrkov, T.V., Avchaciov, K., Tarkhov, A.E., Menshikov, L.I., Gudkov, A.V., & Fedichev, P.O. (2021). Longitudinal analysis of blood markers reveals progressive loss of resilience and predicts human lifespan limit. *Nature Communications*, 12, 2765.

[3] Strogatz, S.H. (2000). From Kuramoto to Crawford: exploring the onset of synchronization in populations of coupled oscillators. *Physica D: Nonlinear Phenomena*, 143(1–4), 1–20.

[4] Barzilai, N., Crandall, J.P., Kritchevsky, S.B., & Espeland, M.A. (2016). Metformin as a tool to target aging. *Cell Metabolism*, 23(6), 1060–1065.

[5] Rakshit, K., et al. (2023). Zeitgeber-based interventions and biological aging: a systematic review of exercise timing, sleep regularity, and intermittent fasting in animal and human models. *Ageing Research Reviews*, 85, 101850.

[6] Monk, T.H. (2005). Aging human circadian rhythms: conventional wisdom may not always be right. *Journal of Biological Rhythms*, 20(4), 366–374.

[7] Centers for Disease Control and Prevention. (2020). National Health and Nutrition Examination Survey Data 2017–2018. U.S. Department of Health and Human Services. https://www.cdc.gov/nchs/nhanes/

[8] López-Otín, C., Blasco, M.A., Partridge, L., Serrano, M., & Kroemer, G. (2023). Hallmarks of aging: An expanding universe. *Cell*, 186(2), 243–278.